

SimplexLP

Revised simplex and interior point methods for linear programming

Šehzada Sijarić (19964)

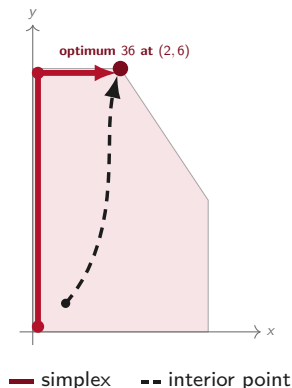
Project #13 | built on the natID SDK

Two ways to solve the same problem

$$\min c^T x \quad \text{s.t.} \quad Ax = b, x \geq 0$$

- **Revised simplex** — walks the **boundary**, vertex to vertex
- **Interior point** — cuts through the **middle**

Both written from scratch, on the **same** natID sparse LU solver, and animated on the same model in the same window.



Dantzig's example: $\max 3x + 5y; \quad x \leq 4, 2y \leq 12,$
 $3x + 2y \leq 18.$

Getting to standard form

`LPPProblem` rewrites the user's model:

- $\max \rightarrow \min$ of $-c$; sign restored for display
- $\leq$ gets a slack (+1), $\geq$ a surplus (−1)
- $x \geq 0$ is implicit

Stored column-wise in `SparseColMatrix` — exactly how the revised simplex prices, and exactly what `addTriple` consumes.

Free starting basis

`slackOfRow[i]` records the +1 slack of row i . Those columns form an **identity submatrix**, so whenever $b \geq 0$ there is a ready-made basis and **phase I is skipped entirely**.

That is the common case for the small models typed into the 2D and 3D tabs.

01

The two solvers

Revised simplex

With basis B and $\mathcal{B} = A(:, B)$, each iteration:

$x_B = \mathcal{B}^{-1}b$	FTRAN
$y = \mathcal{B}^{-T}c_B$	BTRAN
$d_j = c_j - y^T a_j$	reduced costs
$w = \mathcal{B}^{-1}a_q$	FTRAN
$\theta = \min_i \{x_B(i)/w_i : w_i > 0\}$	ratio test

Why revised

Only the **basis** is factorized — the full tableau is never formed. That is what makes a sparse LU the natural partner.

Anti-cycling. Dantzig's rule, switching to **Bland's rule** after a run of degenerate pivots.

Not theoretical: the suite runs **Beale's cycling example**, where plain Dantzig loops forever and this terminates.

Interior point: Mehrotra predictor–corrector

With $D^2 = \text{diag}(x/s)$, $r_p = b - Ax$, $r_d = c - A^T y - s$, each step solves

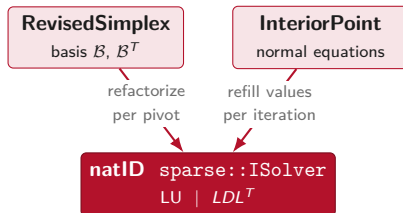
$$(AD^2A^T) \Delta y = r_p + AS^{-1}(Xr_d - r_{xs}), \quad \Delta s = r_d - A^T \Delta y, \quad \Delta x = S^{-1}(-X\Delta s - r_{xs}).$$

- Affine pass: $r_{xs} = XSe$
- Corrector **reuses the same factorization**, with $\sigma = (\mu_{\text{aff}}/\mu)^3$
- $\Rightarrow$ one factorization, **two solves per iteration**
- Steps take 0.99 of the distance to the boundary

Exploiting structure

AD^2A^T is **symmetric positive definite**
 $\Rightarrow$ `Symmetry::SymmetricPosDef`, lower triangle only. The sparsity pattern is computed **once**, symbolically; each iteration refills only the values.

Both solvers sit on the natID sparse LU



`BasisFactor` creates **two** solvers per refactorization — one for $\mathcal{B}$, one for $\mathcal{B}^T$ — because `ISolver` offers no transpose solve and no way to reset values.

Refactorizing every pivot is the robust choice and exercises the SDK on **every** iteration. Forrest–Tomlin updates are future work.

02

The application

What you can actually do with it

2D & 3D tabs

- Type $3x+2y \leq 18$
- Polygon by Sutherland–Hodgman clipping; polytope by enumerating plane triples
- Pivot animation + IPM trajectory from **identical** data

Benchmark tab

- Sweeps synthetic instances of growing size
- Generated **with both feasibility certificates**, so an optimum provably exists
- Runs on a worker thread

MPS & CLP

- Every instance exported as MPS
- `run_clp.sh` re-solves each with COIN-OR CLP
- $\Rightarrow$ compared against an **external solver**

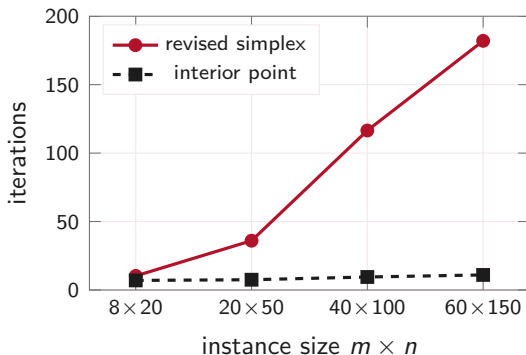
Also

Bilingual (EN / BA) | CMake builds GUI + console tool | installers for all three platforms.

03

Results

Simplex vs. interior point: iteration counts



Mean over the generated instances at each size (8 instances total).

The textbook contrast

Simplex climbs **10** $\rightarrow$ **182**.

Interior point stays at **7–11**.

- All 8 instances **Optimal**
- Objectives agree to 10^{-8}
- KKT residuals pass

Shim LU, so iteration counts are the meaningful figure; the benchmark tab repeats this at scale with the real natID factorizations.

Validation: 86 checks, 0 failures

- Dantzig's example: optimum 36 at (2, 6), zero phase-I iterations
- $A \geq$ model exercising phase I; an equality model
- Infeasible and unbounded **detected explicitly**
- Degenerate vertex; **Beale's cycling example**
- Duplicated equality rows (redundancy)
- MPS write→read round trip, exact
- Exact vertex/face counts: polygon, cube, tetrahedron

Tested exactly as shipped

The suite compiles the **unmodified** solver headers against shims of the SDK interfaces — no test-only modifications.

And the full GUI and console tool are **syntax-checked against the real SDK headers**, so every natID call site is verified independently of the shims.

One foundation, two families of algorithms — and the difference made visible

The revised simplex factorizes only the basis and moves along the boundary; the interior point method factorizes the normal equations and moves through the interior. The same window shows both reaching the same optimum on the same model.

Future work

- Forrest–Tomlin basis updates
- Steepest-edge / Devex pricing
- Harris two-pass ratio test
- Homogeneous self-dual IPM — could also certify infeasibility
- Presolve stage
- Full MPS bounds $\Rightarrow$ read Netlib directly

Thank you

Šehzada Sijarić | Project #13

github.com/ssijaric1/ProjN0_SimplexL_Sijaric